

Product AI Documentation

Clinical Triage AI Architecture, Dual-Agent Consensus, and Safety Guarantees

Document ID: PROD-AI-v2.0 | Date: September 19, 2026 | Clinical Decision Support & Safety Architecture

1. Multi-Agent AI System Architecture

Single-model AI systems in healthcare introduce unacceptable failure modes: hallucinations, missed clinical red flags, and conversational drift. The platform eliminates single-point-of-failure risks through a modular multi-agent pipeline pairing generative LLM reasoning with deterministic rule validation:

2. Dual Independent Assessment & Conservative Consensus

Every completed patient follow-up encounter undergoes simultaneous, independent triage evaluation:

Triage Engine	Evaluation Mechanism	Clinical Role & Invariants
Assessment A (LLM Reasoning)	Gemini 2.5 Flash / OpenAI structured Pydantic extraction	Evaluates conversational nuances, multi-symptom trajectory, emotional distress, and evidence citations from hospital protocol guidelines.
Assessment B (Deterministic Validator)	Hardcoded Python clinical rule validator (regex & keyword tree)	Inflexible safety net. Matches emergency terms (chest pressure, dyspnea, fever > 101, purulent, stopped medication). Cannot be hallucinated away.
Conservative Consensus Arbiter	Invariant Decision Engine	Enforces safety invariant: If EITHER Assessment A or B flags URGENT, final consensus is URGENT. Disagreements mandate human reviewer review.

3. Empirical Clinical Safety Benchmark (30 Cases)

The AI safety architecture was rigorously evaluated against 30 diverse clinical threat cases (covering acute coronary syndrome, surgical site sepsis, pulmonary embolism, medication non-adherence, routine recovery, and prompt injection attacks):

Evaluation Metric	Target Benchmark	Observed Performance	Clinical Verification Status
False Negative Rate (FNR)	< 0.50%	0.00% (0 / 30)	PASSED — Zero Silent Failures
Clinical Sensitivity / Recall	> 98.0%	100.0%	PASSED — All Threats Detected
Adversarial Prompt Defense	100.0%	100.0%	PASSED — Injections Escalated
JSON Schema Conformance	100.0%	100.0%	PASSED — Zero Parsing Errors

4. Controlled AI Tool Layer & EHR Write-Back

AI agents have zero direct database access. Agents interact exclusively through the ControlledToolRegistry. Every tool execution is authenticated against the tenant context, validated against strict Pydantic schemas, and written to the audit log. The EHRSERVICE ensures that progress notes and observation syncs are idempotent, preventing duplicate chart entries upon retries.

Human-in-the-Loop Oversight Desk: When the Conservative Arbiter flags an escalation, automated closing is suspended. The encounter appears in the Clinical Reviewer Inbox with side-by-side assessment diffs and verbatim dialogue quotes. Clinicians record intervention notes and mark resolution, logging an immutable audit event.